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Busway connection through exterior wall

Abstract

A modular data center includes an outer wall having an opening formed therein. The opening is configured to enable a busway to pass through the outer wall. The modular data center further includes an electrical box configured to be coupled to the busway. The electrical box is positioned inboard with respect to the outer wall. The modular data center further includes a shroud configured to extend through the opening of the outer wall. The shroud includes a plurality of outer edges coupled to an outer surface of the outer wall. The modular data center further includes a cover, including a cover opening, configured to enable the busway to pass through the cover opening, and to at least partially seal the opening of the outer wall having the busway passing therethrough.

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Background/Summary

BACKGROUND OF DISCLOSURE

1. Field of Disclosure

(1) This disclosure is directed to busways used in data centers, and more particularly to a system and method of extending a busway through an outer or exterior wall of a facility, such as a data center or a modular data center.

2. Discussion of Related Art

(2) Busway electrical distribution systems are well known in the art of electrical distribution. Busway systems are comprised of a number of factory-assembled sections, each including a number of individually insulated generally flat electrical conductors or busbars stacked one upon another and enclosed within a housing which provides protection and support for the busbars. For example, typical powerbus busway systems have ten-foot (or four-foot) sections of flat, stacked,

electrical conductors for transporting electrical energy from a point A to a point B, while distributing the electrical energy to various electrical loads. Busway construction is modular, and, in many ways, is superior to cable and conduit systems from an installation standpoint.

(3) Typically, a busway can be used internally within a facility or externally to supply power to a facility, such as a data center or a modular data center. Although the busway can pass through a closed wall section of the facility, the busway is not typically terminated at the interface. Further, the busway is typically not rated to go through a closed wall section of the facility.

SUMMARY OF DISCLOSURE

(4) One aspect of the present disclosure is directed to a modular data center comprising an outer wall having an opening formed therein. The opening is configured to enable a busway to pass through the outer wall. The modular data center further comprises an electrical box configured to be coupled to the busway. The electrical box is positioned inboard with respect to the outer wall. The modular data center further comprises a shroud configured to extend through the opening of the outer wall. The shroud includes a plurality of outer edges coupled to an outer surface of the outer wall. The modular data center further comprises a cover, including a cover opening, configured to enable the busway to pass through the cover opening, and to at least partially seal the opening of the outer wall having the busway passing therethrough.

(5) Embodiments of the modular data center further may include configuring the shroud with a plurality of walls that extend in a direction generally perpendicular to the plurality of outer edges. The plurality of outer edges may be configured to extend in an outboard direction with respect to the outer wall. A bottom wall of the plurality of walls of the shroud may be sloped downwardly toward the plurality of outer edges. Inner surfaces of the plurality of outer edges each may include a seal configured to engage the outer surface of the outer wall. The shroud further may include a plurality of inner edges configured to be coupled to the electrical box. The plurality of inner edges may be configured to extend in an inboard direction with respect to the outer wall. The cover may be configured to be secured to the plurality of outer edges of the shroud. The cover opening may be sized to receive the busway. The cover opening may include a gasket to create at least a partial seal between the busway and the cover. The cover further may include at least one slider plate mounted on the cover adjacent an edge of the opening. The at least one slider plate may be configured to improve the seal between the busway and the cover. The cover may include one or more small openings formed along a bottom edge of the cover. The modular data center further may include a shield secured to the outer wall and positioned over the busway. The shield may include a top wall and two side walls extending downwardly from the top wall. The modular data center further may include at least one heater secured to a wall of the shroud. The modular data center further may include a thermostat coupled to the at least one heater, the thermostat being configured to control the at least one heater. The at least one heater may include two heaters. The modular data center further may include a ceiling coupled to the outer wall, with the ceiling including at least one support member configured to support the electrical box. The modular data center further may include a clamp configured to secure the electrical box, with the clamp being coupled to the at least one support member to enable the electrical box slide with respect to the ceiling.

(6) Another aspect of the present disclosure is directed to an assembly to protect a busway configured to be coupled to an electrical box, with the busway extending through an opening of an outer wall of a modular data center. In one embodiment, the assembly comprises a shroud configured to extend through the opening of the outer wall. The shroud includes a plurality of outer edges coupled to an outer surface of the outer wall. The assembly further comprises a cover, including a cover opening, configured to enable the busway to pass through the cover opening, and to at least partially seal the opening of the outer wall having the busway extending therethrough.

(7) Embodiments of the assembly further may include configuring the shroud with a plurality of walls that extend in a direction generally perpendicular to the plurality of outer edges. The plurality of outer edges may be configured to extend in an outboard direction with respect to the plurality of

walls. A bottom wall of the plurality of walls of the shroud may be sloped downwardly toward the plurality of outer edges. Inner surfaces of the plurality of outer edges each may include a seal configured to engage the outer surface of the outer wall. The shroud further may include a plurality of inner edges configured to be coupled to the electrical box. The plurality of inner edges may be configured to extend in an inboard direction with respect to the plurality of wall. The cover may be configured to be secured to the plurality of outer edges of the shroud. The cover opening may be sized to receive the busway. The cover opening may include a gasket to create at least a partial seal between the busway and the cover. The cover further may include at least one slider plate mounted on the cover adjacent an edge of the opening. The at least one slider plate may be configured to improve the seal between the busway and the cover. The cover may include one or more small openings formed along a bottom edge of the cover. The assembly further may include a shield secured to the outer wall and positioned over the busway. The shield may include a top wall and two side walls extending downwardly from the top wall. The assembly further may include at least one heater secured to a wall of the shroud. The assembly further may include a thermostat coupled to the at least one heater. The thermostat may be configured to control the at least one heater. The at least one heater may include two heaters.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:
- (2) FIG. 1 is a perspective view of a portion of a modular data center of an embodiment of the present disclosure;
 - (3) FIG. 2 is a front view of the modular data center;
 - (4) FIG. 3 is a cross-sectional view of the modular data center taken along line 3-3 in FIG. 2;
 - (5) FIG. 4A is a front perspective view of a shroud of an embodiment of the present disclosure;
 - (6) FIG. 4B is a back perspective view of the shroud shown in FIG. 4A;
 - (7) FIG. 4C is a side view of the shroud shown in FIGS. 4A and 4B;
 - (8) FIG. 5 is a perspective view of a cable tap box;
 - (9) FIG. 6 is a side cross-sectional view of a portion of the modular data center;
 - (10) FIG. 7 is another perspective view of the modular data center looking toward a ceiling and an outer wall of the modular data center;
 - (11) FIG. 8 is a front view of the modular data center showing a completed connection and one with parts removed to reveal the cable tap box;
 - (12) FIG. 9 is a cross-sectional perspective view of the modular data center looking in and down on the cable tap box;
 - (13) FIG. 10 is a side cross-sectional view of a portion of the modular data center showing a heater and a thermostat;
 - (14) FIG. 11 is a perspective view of a portion of the modular data center showing the heaters for each shroud;
 - (15) FIG. 12 is a front view of the heater and the thermostat;
 - (16) FIG. 13A is front perspective view of a top part of an outer cover of an embodiment of the present disclosure;
 - (17) FIG. 13B is a back perspective view of a bottom part of the cover of an embodiment of the present disclosure;
 - (18) FIG. 14 is an enlarged perspective view of a portion of the modular data center showing the

assembled cover and a busway extending through the cover;

(19) FIG. 15 is a front view of the modular data center showing the assembled cover and shields;

(20) FIG. 16 is a back perspective view of a portion of the modular data center showing the installation of the cable tap box and a method of lifting and supporting the cable tap box into position;

(21) FIG. 17 is an enlarged back perspective view of the portion of the modular data center shown in FIG. 16;

(22) FIG. 18 is an enlarged back perspective view of a portion of the modular data center showing the cable tap boxes readied for cable installation;

(23) FIG. 19 is a back perspective view of a portion of the modular data center showing the cable tap boxes each in an installed position;

(24) FIG. 20 is a perspective view of a cable tap box showing removable panels of the cable tap box;

(25) FIG. 21 is another back perspective view of a portion of the modular data center showing rails to install the cable tap box to allow the cable tap box to be slid into position above switchgear;

(26) FIG. 22 is another back perspective view of a portion of the modular data center showing another method of installing the cable tap boxes;

(27) FIG. 23 is an enlarged perspective view of a portion of the modular data center showing hardware used to mount the cable tap box in place;

(28) FIG. 24 is an enlarged perspective view of a framing system and a clamp used to support the cable tap box;

(29) FIG. 25 is a perspective view of the clamp; and

(30) FIG. 26 is a perspective view of a roller assembly.

DETAILED DESCRIPTION

(31) This disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The principles set forth in this disclosure are capable of being provided in other embodiments and of being practiced or of being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(32) Busways may be used to provide power to modules positioned within configurable racks located within the data center. A typical busway includes a sheet metal duct having resin insulated copper or aluminum busbars to conduct a substantial current of electricity. Busways are used in many electrical power distribution devices, such as power modules, switching apparatus, distribution apparatus, and batteries. Oftentimes, busways are connected to busway cable tap boxes, which are used to draw power from the busways. In certain embodiments, the busway cable tap box is mounted on a ceiling and includes a busway splice joint which mates with the busway.

(33) Modular data centers provide additional challenges. Previously, when installers of electronic equipment desired to run a busway to a modular data center, the installer was required to wait until the modular data center was delivered and then create an opening in an exterior wall of the modular data center. Also, the installer needed to create a watertight seal around the opening to ensure water was prohibited from entering the modular data center.

(34) Next, the installer would be required to connect the busway to the equipment inside the modular data center, including a busway cable tap box, thereby requiring the installer to enter the modular data center. Sometimes, customers do not want to provide access to service personnel within the modular data center for a variety of reasons, including the prevention of damage caused by the installer when connecting cables to the equipment within the modular data center. In addition, switchgear can be tested and commissioned at the factory to save time at the customer site.

(35) The present disclosure offers a solution that enables a continuation of a busway through an exterior wall while weatherproofing the opening through which the busway extends. Installers need not access inside the building thereby ensuring the protection of the facility and the cleanliness of the facility while allowing work to continue without the customer needing to be present.

(36) Referring to FIG. 1, a portion of a modular data center is generally indicated at **10**. In one embodiment, the modular data center **10** is a modular structure that is designed to house and support electrical equipment. As shown, the modular data center **10** includes a floor **12**, a wall **14** including several panels, and a ceiling **16**. In some embodiments, the modular data center **10** further includes a variety of electronic modules, e.g., switchgear **18**, supported by the floor **12** of the modular data center. Positioned outside the modular data center **10** are electrical equipment, such as a transformer **20** and a generator **22**. Although two units of electrical equipment **20**, **22** are shown and described herein, any number of units of equipment may be provided based on the size of the modular data center **10**. As shown, a busway **24** is connected to the transformer **20** to provide electrical connection between the transformer **20** and a cable tap box **60** (FIG. 3). Similarly, another busway **24** is connected to the generator **22** to provide electrical connection between the generator and another cable tap box **60** provided within the modular data center **10**. There are openings in the wall **14** to enable the busways **24** to connect the transformer **20** and the generator **22** to their respective cable tap boxes.

(37) Referring additionally to FIGS. 2 and 3, the modular data center **10** further includes an assembly, generally indicated at **30**, that is provided to enable and protect the busway **24** as it passes through an opening **32** provided in the wall **14** of the modular data center. As shown, the assembly **30** includes a shroud, generally indicated at **34**, and a cover, generally indicated at **36**. The shroud **34** functions as a guide to enable the busway **24** to pass through the opening **32** and to connect to the cable tap box **60**. The cover **36** functions to enclose an opening created by the shroud **34** and to seal the opening to prevent unwanted moisture from entering the modular data center **10**. The assembly **30** further includes a shield, generally indicated at **38**, which is secured to the wall **14** of the modular data center **10** above the opening **32**. As its name implies, the shield **38** is provided to divert moisture, e.g., rain, from the opening **32** and the opening created by the shroud **34** and protected by the cover **36** thereby providing additional protection to the busway **24** and its connection to the cable tap box **60** provided within the shroud.

(38) Referring to FIGS. 4A-4C, the shroud **34** is a generally rectangular structure that is configured to extend through the opening **32** of the wall **14** of the modular data center **10**. As shown, the shroud **34** includes four side walls **40a**, **40b**, **40c**, **40d**, each having a relatively substantial width. The width of the side walls **40a-40d** is sufficient to enable the connection of the busway **24** to the cable tap box **60**. In the shown embodiment, side walls **40a**, **40b** are longer than side walls **40c**, **40d** to create a rectangular structure. However, it should be understood that the size and shape of the shroud **34** can be configured to achieve a desired size and shape to accommodate the size and shape of the busway **24** that extends through the opening **32** of the wall **14**. The shorter side walls **40c**, **40d** each can include a relatively small opening **42** formed therein to enable wires, e.g., wires for heater elements, to pass through the shroud **34**.

(39) The shroud **34** further includes four outer edges **44a**, **44b**, **44c**, **44d**, one for each side wall **40a**, **40b**, **40c**, **40d**, respectively, which extend in an outboard direction generally perpendicular to their respective side walls. The outer edges **44a-44d** are welded or bent to their respective side walls **40a-40d** and positioned to engage and secure the shroud **34** to the outer surface of the wall **14**, with the side walls extending into an interior of the modular data center **10**. As shown, the outer edges **44a-44d** of the shroud **34** have a plurality of openings, each indicated at **46**, which are spaced apart from one another to secure the shroud to the wall **14** of the modular data center **10**. In one embodiment, screw fasteners (not shown) can be provided to secure the shroud **34** to the wall **14**. Further, the outer edges **44a-44d** of the shroud **34** have inner surfaces having a seal **48** that is configured to engage the outer surface of the wall **14** to seal the shroud to the wall. In a certain

embodiment, a sealing tape may be applied to the inner surfaces of the outer edges **44a-44d** to seal the shroud **34** to the wall **14**.

(40) The shroud **34** further includes four inner edges **50a, 50b, 50c, 50d**, one for each side wall **40a, 40b, 40c, 40d**, respectively, that extend in an outboard direction generally perpendicular to their respective side walls. The inner edges **50a-50d** are welded or bent to their respective side walls **40a-40d** and provided to engage and secure the shroud **34** to the cable tap box **60**, which is sometimes referred to herein as an electrical box. As with the outer edges **44a-44d** of the shroud **34**, the inner edges **50a-50d** of the shroud **34** have a plurality of openings, each indicated at **52**, that are spaced apart from one another to secure the shroud to the cable tap box **60**. In one embodiment, screw fasteners can be provided to secure the shroud **34** to the cable tap box **60**. Further, the inner edges **50a-50d** of the shroud **34** have outer surfaces having a seal **54** that is configured to engage the outer surface of the cable tap box **60** to seal the shroud to the cable tap box.

(41) In one embodiment, with particular reference to FIG. **4C**, a bottom longer side wall **40b** of the shroud **34** is sloped downwardly from the inner edges **50a-50d** to the outer edges **44a-44d**. When mounted on the wall **14**, the top longer side wall **40a** of the shroud extends generally perpendicularly from the wall **14** whereas the bottom longer side wall **40b** slopes downwardly with respect to a horizontal plane. The subtle slope of the bottom longer side wall **40b** enables fluid, e.g., water, to drain from the shroud **34** should fluid enter the shroud.

(42) Referring to FIG. **5**, an exemplary cable tap box **60** is shown and described herein. As shown, the cable tap box **60** includes a busway interface **62** that is configured to connect the busway **24** to the cable tap box. In one embodiment, the cable tap box **60** is an outdoor cable tap box that is rated for 4000 amps (A). Although oriented in a horizontal fashion, the busway interface **62** of the cable tap box **60** can be oriented vertically based on a desired use. The busway **24** is connected to the busway interface **62** of the cable tap box **60** in a manner shown in FIG. **3**.

(43) Referring to FIGS. **6** and **7**, the cable tap box **60** is secured to the ceiling **16** of the modular data center **10** by clamp assemblies, each generally indicated at **70**, that are configured to slide within a metal framing system **72** provided on the ceiling. In another embodiment, the cable tap box **60** can be placed directly on the switchgear **18**. In the shown embodiment, the framing system includes two tracks **74a, 74b**, which are spaced apart from one another a distance generally corresponding to a width of the cable tap box **60**. The manner in which the cable tap box **60** is secured to the ceiling **16** will be described in detail with reference to FIGS. **24-27**. The clamp assemblies **70** are configured to slide within the tracks **74a, 74b** of the metal framing system **72**, which in one embodiment is sometimes referred to as Unistrut® channels or framing, to enable the cable tap box **60** to be moved from a remote location to an operational position, as shown in FIG. **6**.

(44) In one embodiment, the shield **38** is designed to extend from the wall **14** to divert fluid, e.g., rain, from the busway **24** and busway interface connection **62** to the cable tap box **60**. In one embodiment, the shield **38** is secured to the wall **14** and positioned over the busway **24** and the busway interface **62**, and includes a top wall **76** and two side walls **78, 80** extending downwardly from the top wall (FIG. **2**). A front edge of the top wall **76** of the shield **38** is bent downwardly to further prevent unwanted moisture, e.g., rain, from entering the space below the shield. The top wall **76** of the shield **38** is sized to span beyond a width of the shroud **34** and the cover **38** to protect the busway **24** and the busway interface **62**. This is best shown in FIGS. **1** and **2**.

(45) Referring particularly to FIG. **7**, two cable tap boxes **60** are secured to the ceiling **16** of the modular data center **10** by clamp assemblies **70** configured to slide within tracks **74a, 74b** of two metal framing systems **72** provided on the ceiling. The cable tap boxes **60** are shown to be in their operational positions. Electronic equipment, typically switchgear **18**, are shown to be positioned below the cable tap boxes **60**. It should be understood that any number of cable tap boxes **60** and busways **24** may be provided depending on the size of the modular data center **10** and the number of transformers **20** and generators **22** provided.

(46) Referring to FIG. 8, two assemblies 30 are shown to be provided on the exterior wall 14. The lefthand assembly 30 shows the shroud 34 connected to the outer surface of the exterior wall 14 of the modular data center 10 and to the cable tap box 60. The cable tap box interface 62 is presented to be coupled to the busway 24. The righthand assembly 30 shows the busway 24 connected to the cable tap box 60 and the cover 36 and the shield 38 secured to the shroud 34 and to the exterior wall 14, respectively. As shown, the cover 36 is configured to be secured to the plurality of outer edges of the shroud 34. The cover 36 has a cover opening 94 configured to enable the busway 24 to pass through the cover opening and to at least partially seal the interface of the cover opening and the busway. The cover opening 94 provided in the cover 36 is smaller than the opening 32 of the wall 14 and sized to accommodate the busway. FIG. 9 illustrates the cover 36 providing a seal to the busway 24.

(47) Referring to FIG. 10, the assembly further includes a heating system that is provided to heat the space within the shroud 34 that surrounds the interface 62 between the busway 24 and the cable tap box 60. In one embodiment, the heating system includes two dedicated heaters, each indicated at 100, secured to inner surfaces of the shorter side walls 40c, 40d of the shroud 34. FIG. 10 shows one heater 100 mounted on the inner surface of the shorter side wall 40c of the shroud below the busway interface 62. The other heater 100 is mounted on the inner surface of the opposite shorter side wall 40d of the shroud 34. The heaters 100 are connected to a thermostat 102 that is also mounted on the inner surface of one of the shorter side walls 40c, 40d of the shroud 34, e.g., side wall 40c. The thermostat 102 is provided to control the heaters 100 to achieve a desired temperature within the shroud 34. In one embodiment, the thermostat 102 can be wirelessly controlled so that the assembly 30, when fully assembled, can remain intact. In another embodiment, the thermostat 102 can be controlled by a controller associated with the modular data center 10. The heaters 100 are provided to address moisture within the shroud that surrounds the cable tap box interface connection 62. A humidity sensor can also be provided within the space defined by the shroud 34 as well.

(48) Referring to FIG. 11, each assembly 30 is shown with its respective cover 36 removed. Each assembly 30 is provided with its own heating system that includes the two heaters 100 and two thermostats 102 per assembly. The heating system can be provided for each cable tap box 60 and busway interface 62. Referring to FIG. 12, in one embodiment the heater system includes two ClimaSys 55W heaters and a single ClimaSys CC-simple thermostat, each being provided by Schneider Electric. The heater 100 is a resistance heater and configured to provide localized heat to the space defined by the shroud 34. As shown, the thermostat 102 is suitably connected to the heater 100 (wired or wirelessly) to provide control to the heater.

(49) Referring to FIGS. 13A and 13B, in one embodiment, the cover 36 includes a top part 110 shown in FIG. 13A and a bottom part 112 shown in FIG. 13B. The top part 110 and the bottom part 112, when assembled on the shroud 34, together define the cover opening 94 that is centrally provided within the cover 36. The top part 110 of the cover 36 includes two slider plates, each indicated at 114, which are provided to enclose the sides of the cover opening around the busway 24 once the cover is secured to the shroud 34. When assembled, the cover opening 94 of the cover 36 is sized to seal against the busway 24. The slider plates 114 can be adjusted in a lateral or horizontal direction to adjust a size of the cover opening 94 to engage the busway 24 as it passes through the cover opening. In one embodiment, the top part 110 and the bottom part 112 of the cover 36 each includes a foam gasket 116 surrounding the cover opening 94 to provide a seal or at a minimum at least a partial seal between the cover and the busway 24. The backside of the cover 36 is covered with insulation material 118 to prevent or minimize heat loss within the shroud 34 once the cover is installed.

(50) The top part 110 and the bottom part 112 of the cover 36 further includes several openings, each indicated at 120, provided around the peripheries of the top part and the bottom part of the cover to secure the cover to the shroud 34. In one embodiment, the top part 110 and the bottom part

112 of the cover **36** can be secured by fasteners, such as machine screw or sheet metal screw fasteners. Several drain slots, each indicated at **122**, are provided along the periphery of the bottom part **112** of the cover **36** to serve as drain holes to enable fluid to flow from the cover once the cover is installed on the shroud **34**.

(51) Referring to FIG. **14**, the slider plates **114** of the cover **36** are shown to be mounted on the body of the cover on each side of the cover opening **94** adjacent the cover opening. The slider plates **114** are provided to improve the seal between the busway **24** and the cover **36**. Each slider plate **114** is laterally adjustable to engage the side edge of the busway **24** or busway interface **62** to prevent moisture from entering the shroud **34**. Each slider plate **114**, once adjusted in place, can be locked in place by a lock nut or some other locking device to secure the slider plate against the busway **24** or the busway interface **62**. The drain slots **122** are located below the busway **24** adjacent to a bottom edge of the bottom part **112** of the cover **36**. The openings **120** along the periphery of the top part **110** and the bottom part **112** of the cover **36** can be shared with the shroud **34** to secure the shroud and the cover in place with the fasteners.

(52) Referring to FIG. **15**, the covers **36** are mounted on the shroud **34** to illustrate the completed assembly. In one embodiment the lefthand assembly protects the busway **24** that is provided to connect the transformer **20** to the cable tap box **60** and the righthand assembly protects the busway that is provided to connect the generator **22** to the cable tap box. The rain shields **38** are provided to protect the busways **24** and the interfaces **62** of the busways. It should be understood that the busways **24** can be used to connect any type of electrical equipment, and the provision of the transformer **20** and the generator **22** are exemplary only.

(53) Referring to FIGS. **16-19**, the manner in which the cable tap boxes **60** are installed is shown and described below. As shown in FIG. **16**, the modular data center **10** includes several switchgear modules **18** aligned in a row along the wall **14** of the modular data center. One cable tap box **60** is shown to be fully installed above the switchgear modules **18** in a position in which the cable tap box is adjacent to the wall **14**. As shown in FIGS. **16** and **17**, the other cable tap box **60** is shown in a pre-installation position in which the cable tap box is supported by an equipment jack **130** that is provided on a roller base **132**. The equipment jack **130** is configured to lift the cable tap box **60** toward the ceiling **16** for installation, while the roller base **132** is provided to laterally move or roll the equipment jack and the cable tap box to its desired, operational position. The cable tap box **60** is positioned within an aisle of the modular data center **10** behind the row of switchgear modules **18** and secured by suitable hardware, e.g., clamp assemblies **70**, to the tracks **74a**, **74b** of the framing system **72** provided on the ceiling **16**. As shown in FIGS. **18** and **19**, the cable tap box **60** is moved into the fully installed position within the framing system **72** and suitable clamped in place against the opening **32** in the wall **14** that receives the shroud **34**. Each cable tap box **60** includes cables **136** that are provided to connect the cable tap boxes to the switchgear **18**.

(54) Referring to FIG. **20**, a modified cable tap box is indicated at **140**, the modified cable tap box having a side panel **142** provided on both sides of the cable tap box. Each side panel **142** is configured to be removed by removing fasteners used to secure the side panel to a frame of the cable tap box. A back panel **144** is configured to be removed as well. The removable side panels **142** and the back panel **144** are provided to easily access the interior of the cable tap box **140** after the cable tap box is installed in its operable position.

(55) Referring to FIG. **21**, when installing the cable tap boxes **60**, a gap indicated at **150** can be provided adjacent to a wall **152** of the modular data center **10** and an outer edge of the cable tap box. This gap **150** can enable accessories, hardware and cabling to be installed adjacent to the cable tap box **60**. In one embodiment, the gap **150** is 30-inches wide, but any size gap can be provided. As shown, the switchgear **18** similarly are positioned in the modular data center **10** a distance from the wall **152** that is the same distance as the gap **150** between the wall and the cable tap box.

(56) Referring to FIG. **22**, in one embodiment, a modified framing system **160** can be provided. As shown, the modified framing system generally indicated at **160** is angled to account for a small gap

provided between the cable tap box **60** and the outer edge of the equipment rack **18** and the wall **152** of the modular data center **10**. As shown, a first portion **160a** of the framing system **160** is relatively perpendicular to the wall adjacent to the wall **14** to position the cable tap box **60** in place once fully installed. However, a second portion **160b** of the framing system **160** is angled with respect to the first portion **160a** to enable better access to the cable tap box **60** when the cable tap box is in the pre-installation position. Specifically, in one embodiment, angled framing system **160** can be provided to achieve a small gap, e.g., six inches, from the wall **152** to enable the manipulation and access to the cable tap box **60**.

(57) Referring to FIGS. **23** and **24**, in one embodiment, the cable tap box **60** is secured to the ceiling **16** by the aforementioned clamp assemblies and framing system **72**. The clamp assemblies **70** are configured to clamp onto the cable tap box **60** to support the weight of the cable tap box. As shown in FIG. **23**, four clamp assemblies **70** can be used to secure the cable tap box **60**, with a clamp assembly being provided at each corner of the cable tap box. The weight of the cable tap box **60** can be somewhat substantial, e.g., 400+ pounds (lbs). Thus, the ceiling **16** and each clamp assembly **70** must be sufficiently robust to carry its portion of the weight of the cable tap box **60**. Moreover, the tracks **74a**, **74b** of the framing system **72** must be robust enough to support the weight of the clamp assemblies **70** and the cable tap box **72**. The clamp assemblies **70** further are configured to be coupled to the tracks **74a**, **74b** of the framing system **72** to enable the cable tap box **60** to ride along a length of the framing system between the aforementioned pre-installation and installed positions.

(58) Referring additionally to FIGS. **25** and **26**, each clamp assembly **70** includes a clamp **180** (FIG. **25**) that is secured to the track or rail of the framing system **72** (which is greased) or is secured to a roller assembly **182** (FIG. **26**). In one embodiment, the clamp **180** is a C-clamp in which a throat of the clamp can be widened to enable a flange **184** of the cable tap box **60** to be inserted within the clamp. The clamp **180** is tightened to firmly secure the clamp to the flange **184** of the cable tap box **60**. The manner in which the clamp **180** is secured to the cable tap box **60** is shown in FIG. **24**. Each clamp **180** can be locked in place by a nut **186** provided on the clamp. Alternatively, the roller assembly **182**, which includes four rollers or wheels, each indicated at **188**, can be configured to ride within a channel of the track **74a** or **74b** of the framing system **72**. The roller assembly **182** further includes a bracket **190** that can be secured to the clamp **180**. Another nut **192** is provided on the clamp **180** to firmly secure the clamp to the cable tap box **60**. Once assembled, the clamp hardware or the roller assembly **182** is configured to slide or roll along a length of the track **74a** or **74b** of the framing system **72** while the clamp **180** is secured to the cable tap box **60** to move the cable tap box to a desired location.

(59) A method of assembling a busway cable assembly to protect a busway configured to be coupled to an electrical box is further disclosed. As discussed above, the busway transition of the cable tap box extends through an opening of an outer wall of a modular data center, and is provided to connect equipment within the modular data center, e.g., switchgear, to a transformer or a generator. In one embodiment, the method includes installing a shroud on the wall with the shroud extending through the opening of the wall. The shroud includes a plurality of outer edges that are secured to an outer surface of the outer wall and a plurality of inner edges that are secured to the cable tap box. Inner surfaces of the plurality of outer edges each include a seal configured to engage the outer surface of the outer wall.

(60) The method further includes installing a cover to the shroud. The cover includes a cover opening that is sized to enable the busway to pass through the cover opening. The cover is provided to at least partially seal the opening of the outer wall having the busway extending through. Specifically, the cover is secured to the plurality of outer edges of the shroud and can include a gasket to create at least a partial seal between the busway and the cover. Further, the cover can include slider plates at each end of the cover opening to improve the seal between the busway and the cover.

(61) The method further includes installing a shield to the outer wall, with the shield being positioned over the busway. The shield protects the connection of the busway to the cable tap box interface. The method further includes installing at least one heater to a wall of the shroud and a thermostat, coupled to the heater, to the wall or another wall of the shroud. The thermostat controls the heater.

(62) Thus, it should be understood that the assembly described herein enables a busway to pass through an exterior wall of a modular data center to allow contractors to continue the busway connections outside the facility, for example, to a generator or a transformer, without the need to enter a locked facility. The system allows a busway connection through a wall, behind access panels, to permit system checkout in advance and keep contractors from requiring access inside a locked facility.

(63) Having thus described several aspects of at least one embodiment of this disclosure, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this disclosure, and are intended to be within the spirit and scope of the disclosure. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A modular data center comprising: an outer wall having an opening formed therein, the opening being configured to enable a busway to pass through the outer wall; an electrical box configured to be coupled to the busway, the electrical box being positioned inboard with respect to the outer wall; a shroud configured to extend through the opening of the outer wall, the shroud including a plurality of outer edges configured to be coupled to an outer surface of the outer wall; and a cover, including a cover opening, configured to enable the busway to pass through the cover opening, and to at least partially seal the opening of the outer wall having the busway passing therethrough.
2. The modular data center of claim 1, wherein the shroud includes a plurality of walls that extend in a direction generally perpendicular to the plurality of outer edges.
3. The modular data center of claim 2, wherein the plurality of outer edges extends in an outboard direction with respect to the outer wall.
4. The modular data center of claim 2, wherein a bottom wall of the plurality of walls of the shroud is sloped downwardly toward the plurality of outer edges.
5. The modular data center of claim 1, wherein inner surfaces of the plurality of outer edges each include a seal configured to engage the outer surface of the outer wall.
6. The modular data center of claim 1, wherein the shroud further includes a plurality of inner edges configured to be coupled to the electrical box.
7. The modular data center of claim 6, wherein the plurality of inner edges extends in an inboard direction with respect to the outer wall.
8. The modular data center of claim 1, wherein the cover is configured to be secured to the plurality of outer edges of the shroud.
9. The modular data center of claim 1, wherein the cover opening is sized to receive the busway, the cover opening including a gasket to create at least a partial seal between the busway and the cover.
10. The modular data center of claim 9, wherein the cover further includes at least one slider plate mounted on the cover adjacent an edge of the opening, the at least one slider plate being configured to improve the seal between the busway and the cover.
11. The modular data center of claim 1, wherein the cover includes one or more slots formed along a bottom edge of the cover.
12. The modular data center of claim 1, further comprising a shield secured to the outer wall and positioned over the busway.

13. The modular data center of claim 12, wherein the shield includes a top wall and two side walls extending downwardly from the top wall.
 14. The modular data center of claim 1, further comprising at least one heater secured to a wall of the shroud.
 15. The modular data center of claim 14, further comprising a thermostat coupled to the at least one heater, the thermostat being configured to control the at least one heater.
 16. The modular data center of claim 14, wherein the at least one heater includes two heaters.
 17. The modular data center of claim 1, further comprising a ceiling coupled to the outer wall, the ceiling including at least one support member configured to support the electrical box.
 18. The modular data center of claim 17, further comprising a clamp configured to secure the electrical box, the clamp being coupled to the at least one support member to enable the electrical box slide with respect to the ceiling.
 19. An assembly to protect a busway configured to be coupled to an electrical box, the busway extending through an opening of an outer wall of a modular data center, the assembly comprising: a shroud configured to extend through the opening of the outer wall, the shroud including a plurality of outer edges configured to be coupled to an outer surface of the outer wall; and a cover, including a cover opening, configured to enable the busway to pass through the cover opening, and to at least partially seal the opening of the outer wall having the busway extending therethrough.
 20. The assembly of claim 19, wherein the shroud includes a plurality of walls that extend in a direction generally perpendicular to the plurality of outer edges.
 21. The assembly of claim 20, wherein the plurality of outer edges extends in an outboard direction with respect to the plurality of walls.
 22. The assembly of claim 20, wherein a bottom wall of the plurality of walls of the shroud is sloped downwardly toward the plurality of outer edges.
 23. The assembly of claim 19, wherein inner surfaces of the plurality of outer edges each include a seal configured to engage the outer surface of the outer wall.
 24. The assembly of claim 19, wherein the shroud further includes a plurality of inner edges configured to be coupled to the electrical box.
 25. The assembly of claim 24, wherein the plurality of inner edges extends in an inboard direction with respect to the outer wall.
 26. The assembly of claim 19, wherein the cover is configured to be secured to the plurality of outer edges of the shroud.
 27. The assembly of claim 19, wherein the cover opening is sized to receive the busway, the cover opening including a gasket to create at least a partial seal between the busway and the cover.
 28. The assembly of claim 27, wherein the cover further includes at least one slider plate mounted on the cover adjacent an edge of the opening, the at least one slider plate being configured to improve the seal between the busway and the cover.
 29. The assembly of claim 19, wherein the cover includes one or more slots formed along a bottom edge of the cover.
 30. The assembly of claim 19, further comprising a shield secured to the outer wall and positioned over the busway.
 31. The assembly of claim 30, wherein the shield includes a top wall and two side walls extending downwardly from the top wall.
 32. The assembly of claim 19, further comprising at least one heater secured to a wall of the shroud.
 33. The assembly of claim 32, further comprising a thermostat coupled to the at least one heater, the thermostat being configured to control the at least one heater.
 34. The assembly of claim 32, wherein the at least one heater includes two heaters.
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